RAMCO Institute of Technology, Rajapalayam

Department of Mechanical Engineering

Academic Year 2020 – 2021 (ODD Semester)

Innovative Practice(s) Description

Name of the Faculty	:	Mr. J.Jerold John Britto AP(Sr.Gr.)/Mech
Subject Code and Title	:	OML 751 Testing of Materials
Year and Branch	:	IV year Civil, EEE and ECE
Topics Covered	:	Unit -IV: Material Characterization Testing –Electrical and Magnetic Techniques: Resistivity by Four Probe Method
Name of the Laboratory	:	Solid State Physics Virtual Lab
Website Link	:	https://vlab.amrita.edu/?sub=1&brch=282&sim=1512&cnt=1
Date	:	19.10.2020
No.of Students participated	:	61
Assignment given if any (If Yes share the uploaded folder link)	:	-

Demonstration of Material Characterization :Testing-Electrical and Magnetic Techniques

OBJECTIVE

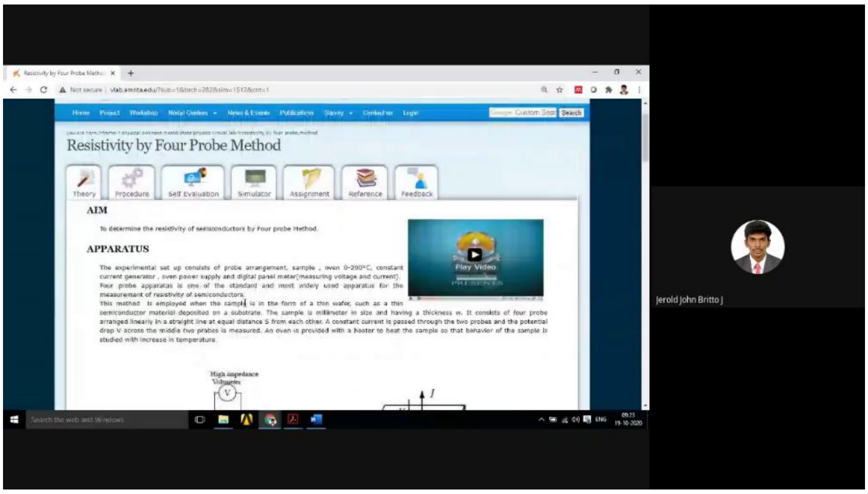
To determine the resistivity of semiconductors by Four probe Method.

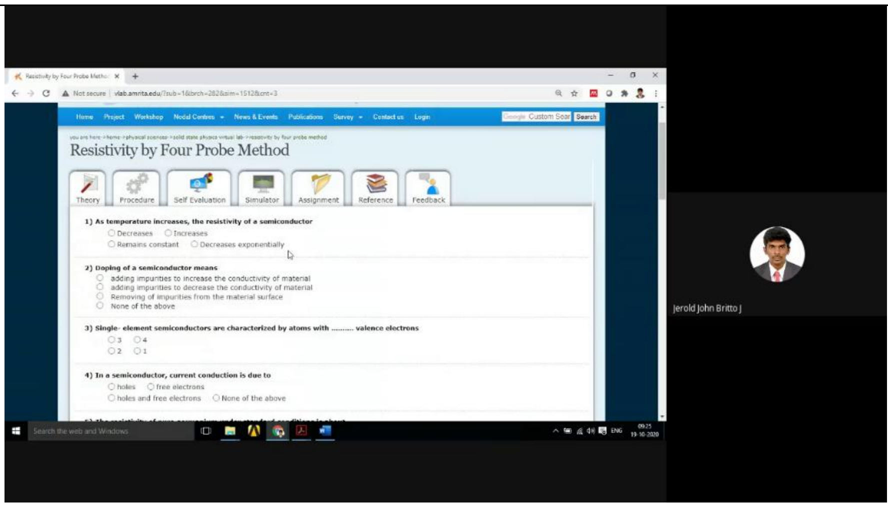
A semiconductor has electrical conductivity intermediate in magnitude between that of a conductor and insulator. Semiconductor differs from metals in their characteristic property of decreasing electrical resistivity with increasing temperature.

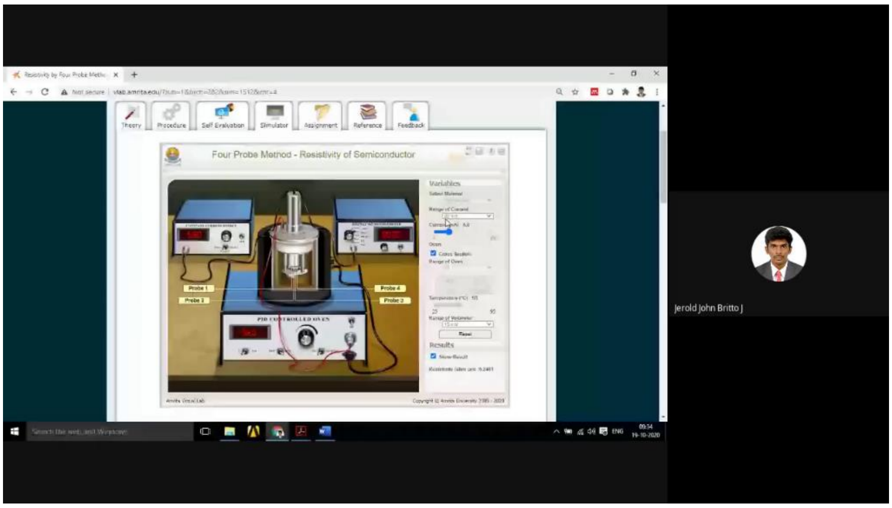
According to band theory, the energy levels of semiconductors can be grouped into two bands, valence band and the conduction band. In the presence of an external electric field, it is electrons in the valence band that can move freely, thereby responsible for the electrical conductivity of semiconductors.

The students can able to know about the resistivity of four probe method and also they can able to select the material, range of current, current' slider, range of oven, selection of range combo box and temperature slider and voltmeter combo box.

Course Outcome / Programme Outcomes	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO8**	**PO9**	**PO10**	**PO12**
CO2 Analyze engineering components using various mechanical testing procedure.	2	2	2	1	2	1	2	1	2




Resistivity by Four Probe Method
Theory
Procedure
Self Evaluation
Simulator
Assignment
Reference
Feedback
1) As temperature increases, the resistivity of a semiconductor
Decreases
Increases
Remains constant
Decreases exponentially
2) Doping of a semiconductor means
adding impurities to increase the conductivity of material
adding impurities to decrease the conductivity of material
Removing of impurities from the material surface
None of the above
3) Single- element semiconductors are characterized by atoms with .... valence electrons
3
4
2
1
4) In a semiconductor, current conduction is due to
holes
free electrons
holes and free electrons
None of the above
Jerold John Britto J


Four Probe Method - Resistivity of Semiconductor
Probe 1
Probe 2
Probe 4
Probe 3
PID CONTROLLED OVEN
Variables
Results
Reset
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